

School of Computer Science and Engineering
Experiment List for Programming Ability and Logic Building - 1

Proposed Date	Le t u r e	Experiment	In Class / Take Home
2/26) W e e k : (16/02/21/0	1	Given a integer matrix (or 2D array) a[][] of dimensions n * m. Also, given another 2- D array query[][] of dimensions q * 4. For each index $0 < i < \text{query.length}$, find the sum of all the elements of the rectangular matrix whose top left corner is (query[i][0], query[i][1]) and bottom right corner is (query[i][2], query[i][3]). Example - Input: n = 3, m = 3, q = 2 a[][] = [[1, 2, 3], [4, 5, 6], [7, 8, 9]] query[][] = [[0, 0, 2, 2] [1, 0, 2, 1]] Output: 45 24 Explanation: The sum of all the elements in the matrix whose top left corner is (0, 0) and the bottom right corner is (2, 2) is 45. The sum of all the elements in the matrix whose top left corner is (1, 0) and the bottom right corner is (2, 1) is 24. Link: https://www.geeksforgeeks.org/problems/prefix-sum-of-matrix-or-2d-array/1?page=1&category=Arrays&sortBy=latest	In Class

	1	Given a matrix a of size n*m which represents a park, there is some construction work needs to be done. You are also given q queries each query contains two numbers R and C, For every query we need to construct a footpath in the Rth row and Cth column, there is a cost of this construction, after the construction this path will divide the park into sections, and the cost of the construction is the sum of minimum value present in all the sections. You are asked to find this cost for all the queries. Note: Elements present in queries array are according to 1-based indexing. Example 1: Input: n=3 m=3 a={{1,2,3},{4,5,6},{7,8,9}}	In Class
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	q=1 queries={{2,2}} Output: 20 Explanation: For query {2,2}, after Footpath construction, park looks like: <pre> 1 * 3 * * * 7 * 9 </pre> Here star represents footpath, Sum of minimum value present in all the remaining sections is 1+3+7+9=20 Example 2: Input: n=3 m=4 a={{1,2,3,4},{5,6,7,8},{1,2,3,4}} q=1 queries={{3,4}} Output: 1 Explantion: For Query {3,4}, after Footpath construction, park looks like: <pre> 1 2 3 * 5 6 7 * * * * * </pre> There is only one section, sum of minimum value present in all the remaining sections is 1. Link: https://www.geeksforgeeks.org/problems/footpath-construction/1?page=1&category=Arrays&sortBy=latest	
1	Given an array arr[], find the minimum number of operations required to make the sum of its elements less than or equal to half of the original sum. In one operation, you may replace any element with half of its value (with floating-point precision). Examples: Input: arr[] = [8, 6, 2] Output: 3 Explanation: Initial sum = (8 + 6 + 2) = 16, half = 8 Halve 8 → arr[] = [4, 6, 2], sum = 12 (still 12 > 8)	Take Home

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		Halve 6 $\rightarrow$ arr[] = [4, 3, 2], sum = 9 (still $9 > 8$) Halve 2 $\rightarrow$ arr[] = [4, 3, 1], sum = 8. Input: arr[] = [9, 1, 2] Output: 2 Explanation: Initial sum = 12, half = 6 Halve 9 $\rightarrow$ arr[] = [4.5, 1, 2], sum = 7.5 (still > 6) Halve 4.5 $\rightarrow$ arr[] = [2.25, 1, 2], sum = $5.25 \leq 6$ Link: https://www.geeksforgeeks.org/problems/minimum-steps-to-halve-sum/1?page=1&category=Arrays&sortBy=latest	
	1	You are given an array arr[] of size n, where arr[i] denotes the range of working hours a person at position i can cover. • If arr[i] $\neq$ -1, the person at index i can work and cover the time interval [i - arr[i], i + arr[i]]. • If arr[i] = -1, the person is unavailable and cannot cover any time. The task is to find the minimum number of people required to cover the entire working day from 0 to n - 1. If it is not possible to fully cover the day, return -1. Examples: Input: arr[] = [1, 2, 1, 0] Output: 1 Explanation: The person at index 1 can cover the interval [-1, 3]. After adjusting to valid bounds, this becomes [0, 3], which fully covers the entire working day 0 to n - 1. Therefore, only 1 person is required to cover the whole day. Input: arr[] = [2, 3, 4, -1, 2, 0, 0, -1, 0] Output: -1 Explanation: Persons up to index 2 cover interval [0...6], but working hour 7 cannot be cover as arr[7] = -1, Since the 7th hour cannot be covered by any person, it is impossible to cover the full working day. Input: arr[] = [0, 1, 0, -1] Output: -1 Explanation: The last hour cannot be covered by any person, so it is impossible to cover the full working day. Link: https://www.geeksforgeeks.org/problems/minimum-sprinklers/1?page=1&category=Arrays&sortBy=latest	Take Home

	1	ou are given an integer array arr[]. For every element in the array, your task is to determine its Previous Smaller Element (PSE). The Previous Smaller Element (PSE) of an element x is the first element that appears to the left of x in the array and is strictly smaller than x. Note: If no such element exists, assign -1 as the PSE for that position. Examples: Input: arr[] = [1, 6, 2] Output: [-1, 1, 1] Explanation: For 1, there is no element on the left, so answer is -1.	Take Home
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		For 6, previous smaller element is 1. For 2, previous smaller element is 1. Input: arr[] = [1, 5, 0, 3, 4, 5] Output: [-1, 1, -1, 0, 3, 4] Explanation: For 1, no element on the left, so answer is -1. For 5, previous smaller element is 1. For 0, no element on the left smaller than 0, so answer is -1. For 3, previous smaller element is 0. For 4, previous smaller element is 3. For 5, previous smaller element is 4. Link: https://www.geeksforgeeks.org/problems/previous-smaller-element/1?page=1&category=Arrays&sortBy=latest	

	1	You are given an integer array arr[]. For every element in the array, your task is to determine its Previous Greater Element (PGE). The Previous Greater Element (PGE) of an element x is the first element that appears to the left of x in the array and is strictly greater than x. Note: If no such element exists, assign -1 as the PGE for that position. Examples: Input: arr[] = [10, 4, 2, 20, 40, 12, 30] Output: [-1, 10, 4, -1, -1, 40, 40] Explanation: For 10, no elements on the left, so answer is -1. For 4, previous greater element is 10. For 2, previous greater element is 4. For 20, no element on the left greater than 20, so answer is -1. For 40, no element on the left greater than 40, so answer is -1. For 12, previous greater element is 40. For 30, previous greater element is 40. Input: arr[] = [10, 20, 30, 40] Output: [-1, -1, -1, -1] Explanation: Each element of the array has no previous greater element. Link: https://www.geeksforgeeks.org/problems/previous-greater-element/1?page=1&category=Arrays&sortBy=latest	Take Home
	1	You are given an integer array arr[]. Your task is to count the number of subarrays where the first element is the minimum element of that subarray. Note: A subarray is valid if its first element is not greater than any other element in that subarray. Examples: Input: arr[] = [1, 2, 1] Output: 5 Explanation: All possible subarrays are:	Take Home

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		[1], [1, 2], [1, 2, 1], [2], [2, 1], [1] Valid subarrays are: [1], [1, 2], [1, 2, 1], [2], [1] -> total 5 Input: arr[] = [1, 3, 5, 2] Output: 8 Explanation: Valid subarrays are: [1], [1, 3], [1, 3, 5], [1, 3, 5, 2], [3], [3, 5], [5], [2] -> total 8 Link: https://www.geeksforgeeks.org/problems/subarrays-with-first-element-minimum/1?page=1&category=Arrays&sortBy=latest	
2/2 6) W ee k : (1 6/ 02 /2 6	2	You are given an array arr[]. The task is to determine whether the array contains a 132 pattern, i.e., three indices i, j and k such that $i < j < k$, arr[i] < arr[j] > arr[k] and arr[i] < arr[k]. Return true if such a triplet exists, otherwise return false. Examples: Input: arr[] = [4, 7, 11, 5, 13, 2] Output: true Explanation: Triplet [4, 7, 5] satisfies the condition since $4 < 7$, $5 < 7$ and $4 < 5$. Input: arr[] = [11, 11, 12, 9] Output: false Explanation: No triplet satisfies the given conditions. Link: https://www.geeksforgeeks.org/problems/find-132-pattern-in-array/1?page=1&category=Arrays&sortBy=latest	In Class
to 21 /0	2	You are given an array arr[], where arr[i] represents the height of the ith person standing in a line. A person i can see another person j if: • height[j] < height[i], • There is no person k standing between them such that height[k] ≥ height[i]. Each person can see in both directions (front and back). Your task is to find the maximum number of people that any person can see (including themselves). Examples: Input: arr[] = [6, 2, 5, 4, 5, 1, 6] Output: 6 Explanation: Person 1 (height = 6) can see five other people at following positions (2, 3, 4, 5, 6) in addition to himself, i.e. total 6. Person 2 (height: 2) can see only himself. Person 3 (height = 5) is able to see people 2nd, 3rd, and 4th person. Person 4 (height = 4) can see himself. Person 5 (height = 5) can see people 4th, 5th, and 6th. Person 6 (height = 1) can only see himself. Person 7 (height = 6) can see 2nd, 3rd, 4th, 5th, 6th, and 7th people. A maximum of six people can be seen by Person 1, 7th	In Class

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		Input: arr[] = [1, 3, 6, 4] Output: 4 Explanation: Person with height 6 can see persons with heights 1, 3 on the left and 4 on the right, along with himself, giving a total of 4. Link: https://www.geeksforgeeks.org/problems/maximum-people-visible-in-a-line/1?page=1&category=Arrays&sortBy=latest	
	2	Given two integers n and k, the task is to find all valid combinations of k numbers that adds up to n based on the following conditions: • Only numbers from the range [1, 9] used. • Each number can only be used at most once. Note: You can return the combinations in any order, the driver code will print them in sorted order. Examples: Input: n = 9, k = 3 Output: [[1, 2, 6], [1, 3, 5], [2, 3, 4]] Explanation: There are three valid combinations of 3 numbers that sum to 9: [1, 2, 6], [1, 3, 5] and [2, 3, 4]. Input: n = 3, k = 3 Output: [] Explanation: It is not possible to pick 3 distinct numbers from 1 to 9 that sum to 3, so no valid combinations exist. Link: https://www.geeksforgeeks.org/problems/comination-sum-iii--111703/1?page=1&category=Arrays&sortBy=latest	Take Home

	2	Given an array arr[] of size n, your task is to divide the array in two subsets such that the absolute difference between the sum of elements in the two subsets is equal to zero (i.e., both subsets have the same sum). • If n is even, both subsets must contain exactly n/2 elements. • If n is odd, one subset must contain (n-1)/2 elements and the other subset must contain (n+1)/2 elements. Note : If multiple answers exist, you may return any of them. The driver code will check and print true if your partition is valid, otherwise false. It is guaranteed that there will always be atleast one valid partition. Examples: Input: arr[] = [1, 2, 3, 4] Output: [[1, 4], [2, 3]] Explanation: The absolute difference between the sum of both subsets is 0. Input: arr[] = [5, 10, 15] Output: [[5, 10], [15]] Explanation: The absolute difference between the sum of both subsets is 0. Link: https://www.geeksforgeeks.org/problems/tug-of-war--122234/1?page=1&category=Arrays&sortBy=latest	Take Home
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	2	Given an array arr[] of integers and an integer k, select k elements from the array such that the minimum absolute difference between any two of the selected elements is maximized. Return this maximum possible minimum difference. Examples: Input: arr[] = [2, 6, 2, 5], k = 3 Output: 1 Explanation: 3 elements out of 4 elements are to be selected with a minimum difference as large as possible. Selecting 2, 2, 5 will result in minimum difference as 0. Selecting 2, 5, 6 will result in minimum difference as 6 - 5 = 1. Input: arr[] = [1, 4, 9, 0, 2, 13, 3], k = 4 Output: 4 Explanation: Selecting 0, 4, 9, 13 will result in minimum difference of 4, which is the largest minimum difference possible. Link: https://www.geeksforgeeks.org/problems/maximize-the-minimum-difference	Take Home

	between-k-elements/1?page=1&category=Arrays&sortBy=latest	
2	Given a sorted array arr[] containing distinct non negative integers that has been rotated at some unknown pivot, and a value x. Your task is to count the number of elements in the array that are less than or equal to x. Examples: Input: arr[] = [4, 5, 8, 1, 3], x = 6 Output: 4 Explanation: 1, 3, 4 and 5 are less than 6, so the count of all elements less than 6 is 4. Input: arr[] = [6, 10, 12, 15, 2, 4, 5], x = 14 Output: 6 Explanation: All elements except 15 are less than 14, so the count of all elements less than 14 is 6. Link: https://www.geeksforgeeks.org/problems/count-elements-less-than-or-equal-to-k-in-a-sorted-rotated-array/1?page=1&category=Arrays&sortBy=latest	Take Home
2	You will be given a vector of integers of size n containing the elements. Your task is to find the sum of all the integers present in the array. As the sum can be large you have to return a value in long long data type. Example 1: Input: n=7 input= {6,2,5,4,5,1,6} Output : 29 Link: https://www.geeksforgeeks.org/problems/add-numbers-in-array/1?page=1&category=Arrays&sortBy=latest	Take Home

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